12-MINUTE READ · 2 WORKSHEETS

Your spreadsheet is *already an app.*

The 10-sign test, the quiet-cost calculator, and a five-move plan to graduate without breaking the business.



BEFORE YOU START

The most important software in your business *was never designed.*

It started as a quick list. Then someone added a tab. Then a formula. Then colour-coding that only Marie understands. Today it's called **Orders_FINAL_v7_REAL.xlsx**, and the business can't run a day without it.

Spreadsheets are brilliant for *thinking*: modelling a price, checking an idea, exploring numbers. They are risky for *running* things: taking orders, tracking jobs, paying people, holding customers' details. The trouble is that the line between the two is crossed quietly, one helpful tab at a time.

This guide helps you see which side of the line you're on, put a dollar figure on it, and move, if you need to, without a big-bang project that stops the business.

86%

of spreadsheets had errors in the field audits that used stronger checking methods.

PANKO, U. OF HAWAIʻI [1]

0.4–2.5%

of formula cells were wrong in real-world audits. On a 2,000-cell sheet, that's 8 to 50 bad cells.

PANKO, FIELD AUDITS [1]

15,841

positive COVID-19 cases missed from England's daily counts in 2020, blamed on a spreadsheet file limit.

PUBLIC HEALTH ENGLAND [3]

How to use this guide (12 minutes)

01 **Take the 10-sign test** on page 3. Be honest; nobody's watching.

02 **Put a price on it** with the quiet-cost calculator on page 6.

03 **Pick your next step** from the three-rung ladder on page 7, and use the column inventory on page 8 to start.

Nothing here says spreadsheets are bad. Most businesses should keep using them. The goal is to stop using them for the jobs they were never built for.

THE 10-SIGN TEST

Is it still a spreadsheet, *or an app in disguise?*

Think of the file your team opens most. Score each sign: **0** = never, **1** = sometimes, **2** = all the time.

1 **Only one person really understands it.** 0 1 2

When they're on holiday, things wait, or someone guesses.

2 **Several people edit it in the same week.** 0 1 2

Somebody's changes get overwritten, or there are two versions and you're not sure which is right.

3 **You copy and paste between files to keep them in sync.** 0 1 2

Order sheet to invoice sheet, schedule to payroll, quote to job list.

4 **Someone re-types data that arrived somewhere else.** 0 1 2

From emails, web forms, PDFs, texts, or paper.

5 **It's hitting a size wall.** 0 1 2

Slow to open, crashes, too big to email, or you've started splitting it by year or by region.

6 **It holds personal information.** 0 1 2

Customers' names, phone numbers, addresses, health details, employee pay. (See guide 05 on Law 25.)

7 **People ask you "what's the status?"** 0 1 2

Customers or staff can't look it up themselves, so they message you and you look it up for them.

8 **You need to know who changed what, and when.** 0 1 2

And the spreadsheet can't really tell you, or the answer is buried in version history nobody checks.

9 **Money decisions come straight out of it.** 0 1 2

Quotes, prices, bids, payroll, stock orders, commissions.

10 **It has "rules" written in colours and capitals.** 0 1 2

"DON'T TOUCH COLUMN H." Yellow means pending. Red means call the client. Hidden tabs nobody dares to delete.

Your total (0 to 20) _____

YOUR SCORE

What your number *is telling you.*

0–5

It's a healthy spreadsheet. Keep it, and tidy it.

It's doing what spreadsheets do best. Spend an hour on the tidy-up list on page 7 so it stays that way, and re-take the test in six months.

6–11

It's become a shared system. Give it a front door.

The sheet is fine for analysis, but people are typing into it from everywhere. The cheapest win is usually a proper form or small portal for the *input*, feeding the sheet you already trust. Re-typing and overwriting stop on day one.

12–20

It's already an app. Just a fragile one.

You're running operations on it: several people, real money, personal data, status questions. It needs logins, roles, a history of changes and rules that can't be broken by a stray paste. Plan the move with the five steps on page 7. It doesn't have to be big or slow.

The two signs that matter most

Whatever your total, look again at **sign 6** (personal information) and **sign 9** (money decisions). A 2 on either one means a single bad paste can cost real money or breach someone's privacy. Those two alone justify a closer look.

> Spreadsheets are a great place for ideas to be born, and a risky place for them to grow old.

A high score isn't a failure. It usually means the business grew faster than its tools. The spreadsheet got promoted to a job nobody wrote a description for.

THREE TRUE STORIES

Careful people. *Overloaded tools.*

2003 · CALGARY

A copy-paste that cost US$24 million

TransAlta, the Alberta power company, bid on New York transmission-hedging contracts using an Excel spreadsheet. During a cut-and-paste, rows were misaligned, so bids landed against the wrong routes. The company won contracts it didn't want, at prices it didn't intend, and took a **US$24 million pre-tax hit**. Its CEO called it "literally a cut-and-paste error in an Excel spreadsheet that we did not detect when we did our final sorting and ranking of bids prior to submission." [2]

2020 · ENGLAND

15,841 positive tests that didn't show up

Lab results arrived as text files and were loaded into spreadsheets. When the files outgrew the limit of the old Excel format, the extra rows were cut off. Between 25 September and 2 October, **15,841 positive cases** were left out of the daily figures and contact tracing was delayed. Public Health England described it as a "file size" issue. [3]

2012 · NEW YORK / LONDON

A formula that divided by the sum instead of the average

JPMorgan's own task force on the "London Whale" losses found a risk model running on spreadsheets fed by copy and paste, a process reviewers called "error prone". One formula divided by the *sum* of two rates instead of their *average*, understating how volatile the positions were. [4]

The pattern behind all three

Manual copying between files. No check to catch a shifted row. A limit nobody knew about. A formula nobody else reviewed. **Each one is a sign on your test.** In a small business the same mistakes show up as a wrong invoice, a missed delivery or a payroll that's off.

WORKSHEET 1 · THE QUIET-COST CALCULATOR

Put a dollar figure *on "it works fine".*

Spreadsheets feel free because nobody gets an invoice for them. Their cost hides in hours and mistakes. Fill in your own column; the example is a fictional 25-person distributor.

COST LINE	EXAMPLE	YOURS
A. Re-typing and copying. Hours a week spent moving data between emails, forms, files. × hourly cost × 48 weeks. 6 h × $40 × 48	$11,520	
B. "What's the status?" Hours a week answering questions people could look up themselves. × hourly cost × 48. 4 h × $40 × 48	$7,680	
C. Fixing mistakes. Errors per month × average cost of one (refund, re-delivery, discount, overtime) × 12. 3 × $250 × 12	$9,000	
D. Waiting on one person. Days a year the key person is away or busy × value of work that stalls each day. 15 days × $200	$3,000	
E. Reconciling. Hours a month checking that two files agree (stock, invoices, payroll). × hourly cost × 12. 5 h × $40 × 12	$2,400	
Quiet cost per year (A + B + C + D + E)	**$33,600**	

Then compare

Ask for a fixed quote to replace just the riskiest part. If the yearly quiet cost is bigger than the quote, the decision has made itself. If it's smaller, keep the sheet and tidy it. Either way, you decided with numbers.

Be conservative

Use a loaded hourly cost (pay plus about 20 to 30% for benefits and overhead) and count only errors you remember. The real figure is almost always higher, because nobody logs the small ones.

YOUR NEXT STEP

Don't jump. *Climb one rung at a time.*

RUNG 1 · SCORE 0–5

Tidy it

- ✓ One **input tab**; every other tab only reads from it.
- ✓ **Drop-down lists** (data validation) instead of free typing.
- ✓ **Lock** formula cells and protect the structure.
- ✓ A **README tab**: what each column means, who owns it.
- ✓ Store it in OneDrive or Google Drive so **version history** works.

RUNG 2 · SCORE 6–11

Add a front door

- ✓ A **web form** for every way data comes in (orders, jobs, requests).
- ✓ Required fields and rules, so bad data can't get in.
- ✓ The form writes into the sheet; analysis stays where it is.
- ✓ Optional: a read-only **status page** for customers or staff.

RUNG 3 · SCORE 12+

Graduate it

- ✓ A small web app with **logins and roles**.
- ✓ A **history of every change**: who, what, when.
- ✓ Rules enforced by the system, not by colours.
- ✓ **Export to Excel** on demand. Your accountant still gets a spreadsheet.

Graduating without breaking the business: five moves

01 **Freeze a copy and write down every column.** What it means, who fills it, where the data comes from. Worksheet 2 on the next page does this for you.

02 **Turn the hidden rules into written rules.** Every colour, comment and "DON'T TOUCH" becomes a sentence: "Orders over $5,000 need a manager's approval." These become the app's validation.

03 **Replace the riskiest tab first, not the whole file.** Usually the one with personal data or money decisions. A small first version, live in weeks, beats a perfect one next year.

04 **Run both side by side for two weeks.** Compare totals every day. Differences are gold: each one is a rule you forgot to write down.

05 **Switch off editing in the old sheet.** Keep it read-only as an archive. Have the new app export the reports people are used to, in the same layout.

WORKSHEET 2 · COLUMN INVENTORY

Map your spreadsheet *in 30 minutes.*

One row per column in your most important tab. This is the single most useful document you can give a developer: half a specification, written in your own words. It also shows you which columns nobody uses any more.

COLUMN	WHAT IT MEANS	WHO FILLS IT	COMES FROM	RULE / ALLOWED VALUES	WHO READS IT
H · Status	Where the order is	Office, drivers	Phone calls	New, Packed, Out, Delivered	Everyone

Hidden rules I found

Colours, comments, "only Marie does this" steps.

Columns nobody uses

Candidates to drop instead of rebuilding.

CHECKLIST · THIS WEEK

Five things to do *before Friday.*

- [ ] **Take the test with a colleague.** Score it separately, then compare. The gaps between your scores are the interesting part.
- [ ] **Make a backup copy** of the file, today, somewhere outside the shared folder. Name it with the date.
- [ ] **Ask "who else knows how this works?"** If the answer is nobody, have the owner record a 10-minute screen video walking through it.
- [ ] **Fill in the quiet-cost calculator** with real numbers from one normal week.
- [ ] **Do the column inventory** for the one tab with money or personal information in it.

Want a second pair of eyes?

Send us your column inventory and quiet-cost number (no real customer data needed: fake names are perfect). An engineer will tell you which rung you're on and, if it's worth it, what replacing the riskiest tab would cost, as a fixed price.

SOURCES

1. Raymond R. Panko, "What We Know About Spreadsheet Errors" (Journal of End User Computing, revised 2008) and "Spreadsheet Errors: What We Know. What We Think We Can Do" (EuSpRIG, arXiv:0802.3457). Field audits with stronger methods found errors in at least 86% of spreadsheets; cell error rates in field audits ranged from about 0.4% to 2.5%.
2. TransAlta Corporation press release, 3 June 2003, filed with the US SEC (Form 6-K); CEO quote reported by The Register, 19 June 2003, and Louise Pryor's spreadsheet-risk blog.
3. Public Health England statement, 4–5 October 2020; reporting by The Register and BBC, 5 October 2020. Accounts differ on the exact row limit; PHE called it a file-size issue.
4. "Report of JPMorgan Chase & Co. Management Task Force Regarding 2012 CIO Losses", 16 January 2013, appendix on the VaR model.

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FROM THE TEAM AT ADDANAPP.COM

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Send a short brief (the template in guide 02 is perfect) and an engineer, not a salesperson, will reply with questions or a fixed quote.

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